

1) Determine the diameter of pipe required

Given:

Kinematic viscosity $\nu = 0.0002 \text{ ft}^2/\text{s}$

Volumetric flow rate $\dot{Q} = 8 \text{ ft}^3/\text{s}$

Frictional head loss $h_f = 0.4 \text{ ft}$

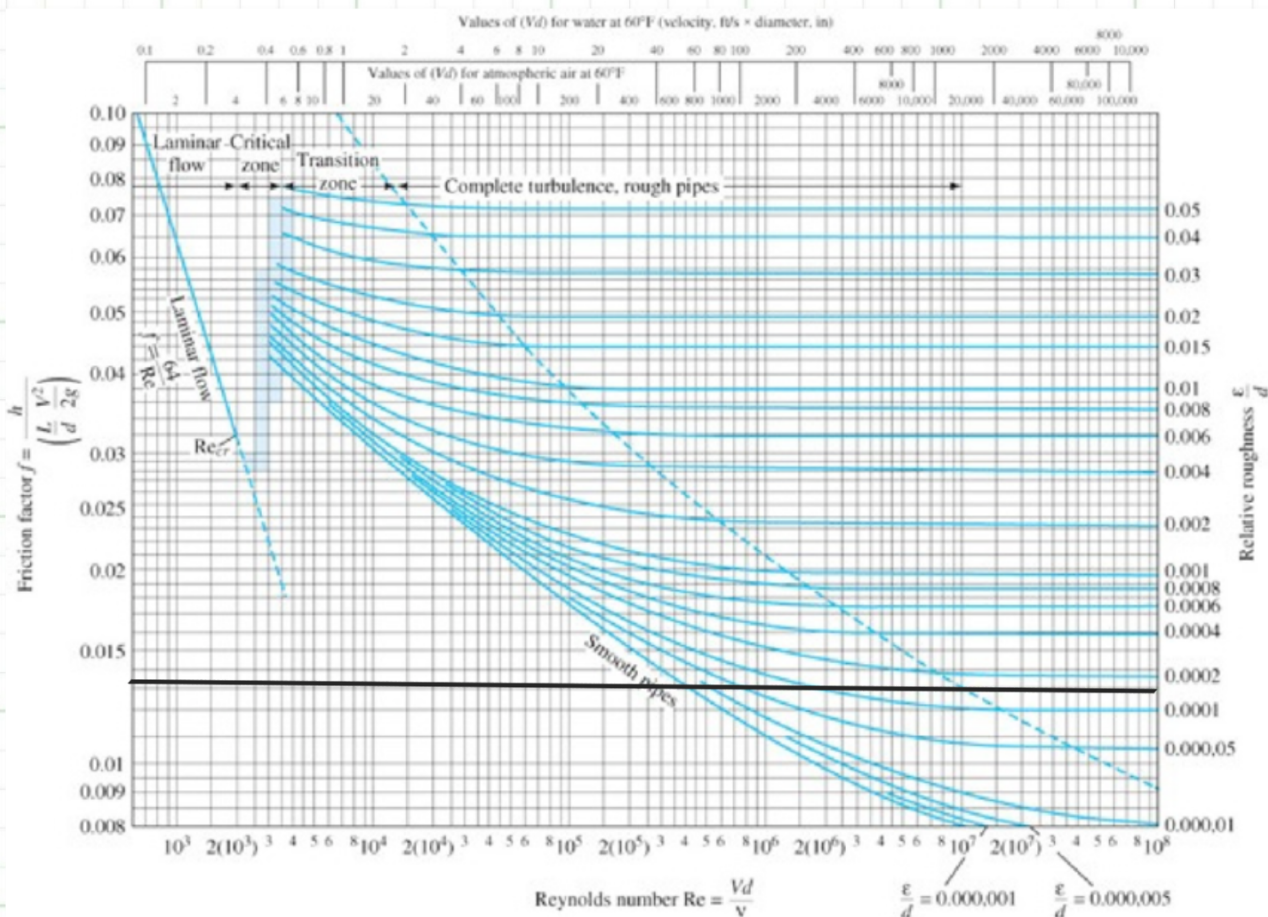
Length of pipe $L = 100 \text{ ft}$

Assume:

$\epsilon = 0.00015 \text{ ft}$ Smooth pipes

Note

- Use the Moody Chart or the Colebrook for iterations



$$\text{Given } \dot{Q} = VA \quad | \quad A = \frac{\pi D^2}{4}$$

$$\Rightarrow V = \frac{4\dot{Q}}{\pi D^2}$$

$$Re = \frac{VD}{\nu} \quad \mu = \nu \cdot \rho$$

$$\Rightarrow Re = \frac{V \cdot D}{\nu} = \frac{\frac{4\dot{Q}}{\pi D^2} \cdot D}{\nu}$$

$$Re = \frac{4\dot{Q}}{\nu \pi D}$$

$$h_f = f \left(\frac{L}{D} \frac{V^2}{2g} \right)$$

Assume $\frac{\epsilon}{D} = 1 \Rightarrow \frac{\epsilon}{D} = 0.00015$
 $\therefore$ from Moody's $f \approx 0.013$

$$\Rightarrow h_f = f \times \frac{L}{D} \times \left(\frac{Q}{A} \right)^2 \frac{1}{2g}$$

$$\frac{2h_f g}{f L} = \frac{\left(\frac{Q}{\frac{\pi D^2}{4}} \right)^2}{D} \Rightarrow D = \sqrt[5]{\frac{f L \left(\frac{Q}{\frac{\pi}{4}} \right)^2}{2 h_f g}}$$

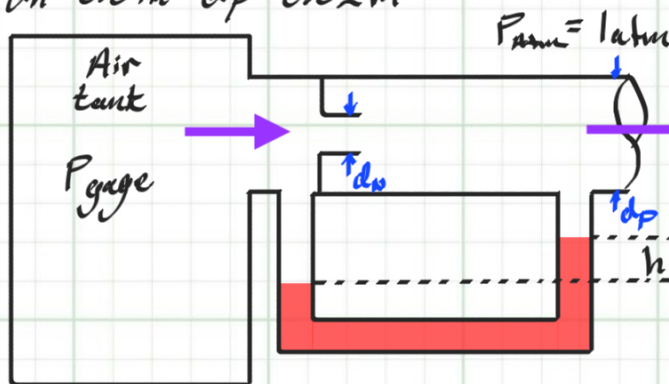
$$D = \sqrt[5]{\frac{(0.013)(100) \left(\frac{4(8)}{\pi} \right)^2}{2(0.4)}} = 2.78 \text{ ft}$$

2) Given:

Tank Pressure P_{gauge} (Pa)	60	320	1200	2050	2470	3500	4900
Manometer Reading h (mm)	6	38	160	295	380	545	820
Friction factor f	0.04059	0.03084	0.02552	0.02345	0.02318	0.02216	0.0214
Volume Flow rate Q_v m ³ /hr (theoretical)	9.05×10^{-2}	0.237	0.499	0.674	0.748	0.908	1.09
Volume Flow rate Q_m m ³ /hr (measured)							

$$\rho_{\text{air}} = 1.2 \text{ kg/m}^3, \mu_{\text{air}} = 0.000015 \text{ kg/ms}, S.G. \text{ oil} = 0.827$$

$$d_n = 0.01 \text{ m}, d_p = 0.02 \text{ m}$$



Assumptions:

- Smooth pipe i.e. roughness is negligible, but Friction Loss is present
- Major and Minor losses
- Minor loss coefficients for entrance $K_x = 0.5$
- Minor loss coefficient for Nozzle $K_n = 0.7$

a) Using the Colebrook Relation fill in above chart's f, Q_v

b) Compute Q_m using Manometer Reading across the Nozzle

c) Plot Flow Rates Vs. the Flow Reynolds # (Re)

$$h_{\text{tot}} = h_f + h_m$$

Using Energy Equation / No heat flux, No pump/turbine work

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + z_2 + h_{\text{tot}}$$

$$*\Delta P = \rho g h_f$$

$$\Rightarrow h_{\text{tot}} = \frac{\Delta P + \rho g \Delta z}{\rho g} \text{ for horizontal flow } \Delta z = 0 \Rightarrow h_{\text{tot}} = \frac{\Delta P}{\rho g}$$

$$\text{Also: } h_{\text{tot}} = \frac{V^2}{2g} \left(\frac{fL}{d} + \sum K \right) \Rightarrow V = \sqrt{\frac{2g h_{\text{tot}}}{\left(\frac{fL}{d} + \sum K \right)}} \Rightarrow V = \sqrt{\frac{2g h_{\text{tot}}}{\left(\frac{fL}{d} + K_n + K_x \right)}}$$

Following the Colebrook Method

Assume Negligible roughness i.e. $\epsilon/d = 0$

$$Re = \frac{\rho V d}{\mu}$$

$$f = \left(\frac{1}{-2 \log_{10} \left(\frac{2.51}{Re} \right)} \right)^2$$

Start: Guess f

↑
No

Find V

Find Re

Converged

Find New f

yes → done $\frac{1}{f_5} = -2 \log_{10} \left(\frac{\epsilon/d}{5.7} + \frac{2.51}{Re} \right)$

$$\epsilon_K = 1.2$$

$$h_{tot} = \frac{P_g}{\rho g}$$

$$V = \sqrt{\frac{2g h_{tot}}{\left(\frac{1}{f} + \epsilon_K\right)}}$$

$$RE = \frac{pVd}{\mu}$$

$$f = \left(\frac{1}{-2 \log_{10} \left(\frac{2.51}{RE} \right)} \right)^2$$

$$\dot{Q}_T (\text{theoretical}) = AV$$

(using V values highlighted)

$$\dot{Q}_T = \frac{\pi D^2}{4} V$$

$$\dot{Q}_T = 7.54 \times 10^{-4} \text{ m}^3/\text{s}$$

$$= 9.05 \times 10^{-2} \text{ m}^3/\text{hr}$$

$$\dot{Q}_T = 1.47 \times 10^{-3} \text{ m}^3/\text{s}$$

$$= 0.237 \text{ m}^3/\text{hr}$$

$$\dot{Q}_T = 4.16 \times 10^{-3} \text{ m}^3/\text{s}$$

$$= 0.499 \text{ m}^3/\text{hr}$$

$$\dot{Q}_T = 5.61 \times 10^{-3} \text{ m}^3/\text{s}$$

$$= 0.674 \text{ m}^3/\text{hr}$$

$$\dot{Q}_T = 6.23 \times 10^{-3} \text{ m}^3/\text{s}$$

$$= 0.748 \text{ m}^3/\text{hr}$$

$$\dot{Q}_T = 7.57 \times 10^{-3} \text{ m}^3/\text{s}$$

$$= 0.908 \text{ m}^3/\text{hr}$$

$$\dot{Q}_T = 9.12 \times 10^{-3} \text{ m}^3/\text{s}$$

$$= 1.09 \text{ m}^3/\text{hr}$$

For Tank (P) T_0 4 significant accuracy

	f	V	RE	new f
@ 60 Pa	0.01	4.383	7016	0.04177
	0.04177	2.363	3781	0.04038
	0.04038	2.401	3841	0.04039
@ 320	f	V	RE	new f
	0.01	10.13	16200	0.03166
	0.03166	6.262	9923	0.03084
@ 1200	0.03084	6.277	10040	0.03084
	f	V	RE	new f
	0.01	14.61	31380	0.02607
@ 2050	0.02607	13.12	20990	0.02551
	0.02551	13.24	21190	0.02552
	f	V	RE	new f
@ 2470	0.01	25.63	41010	0.02421
	0.02421	17.72	28350	0.02374
	0.02374	17.87	28590	0.02375
@ 3500	f	V	RE	new f
	0.01	28.13	45020	0.02361
	0.02361	19.66	31460	0.02317
@ 4900	0.02317	19.83	31730	0.02318
	f	V	RE	new f
	0.01	33.49	53590	0.02255
@ 4900	0.02255	23.89	38220	0.02215
	0.02215	24.08	38230	0.02216
	f	V	RE	new f
@ 4900	0.01	39.63	63410	0.02159
	0.02159	28.81	46100	0.02123
	0.02123	29.03	46440	0.02124

all f converges as shown by

V converge to values shown by

RE Converge to values shown by

using Energy Equation

$$\frac{P_1}{\rho g} = \frac{P_2}{\rho g} + h_m$$

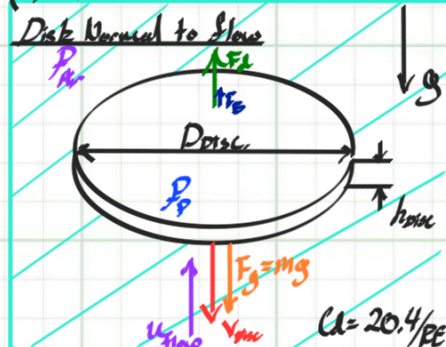
$$h_m = \frac{V^2}{2g} K_m$$

$$\Rightarrow \frac{\Delta P}{\rho g} = \frac{V^2 K_m}{2g}$$

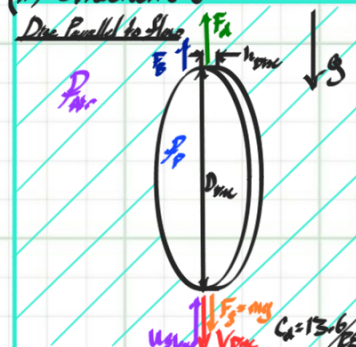
3) Given: $D_{disc} = 25 \mu m$ Thickness $h_{disc} = 5 \mu m$ Density $\rho_p = 1000 \text{ kg/m}^3$ Density $\rho_{air} = 1 \text{ kg/m}^3$
 $\mu_{air} = 0.000015 \text{ kg/ms}$

Determine Terminal Velocity of disc in the two orientations shown:

(i) Orientation



(ii) Orientation



Known:

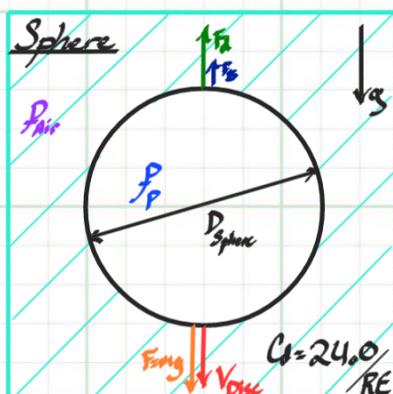
$$C_{d, disc} = \frac{F_D}{\frac{1}{2} \rho V_{disc}^2 A_p}$$

$$A_p = \frac{\pi D_{disc}^2}{4}$$

* Drag coefficient C_d is based on A_p for both orientations *

Neglect Buoyancy.

$$Re = \frac{\rho V D}{\mu}$$



A) Express Newton's law of motion for orientations (i) (ii)

B) Find Symbolic Expression for terminal velocity for "

C) Find the Numerical values for terminal velocity for "

D) Find Symbolic expression for terminal velocity for spherical particle

(Where drag coefficient $C_{d, sphere}$ the frontal Area of Sphere with D_{sphere})

Find Numerical Values for Sphere that will result in $U_{sphere} = U_{disc} (0.4015)$

Newton's laws of Motion

(i) $F_{net} = m_{disc} a = \Sigma F = F_D - F_g$
 $= m_{disc} \frac{dV}{dt} = F_D - F_g$

Symbolic

$$m_{disc} = \rho_p V_{disc} \quad F_g = \rho_p V_{disc} g \quad F_D = C_{d, disc} \cdot \frac{1}{2} \rho_{air} V_{disc}^2 A_p$$

Assume @ low speeds, $Re = 1$ $C_{d, disc} = 20.4/Re \Rightarrow 20.4$

@ Terminal Velocity $\frac{dV}{dt} = 0 \Rightarrow$ Forces in Balance

$$\cancel{\rho_p V_{disc}} \cdot \frac{dV}{dt} = C_{d, disc} \cdot \frac{1}{2} \rho_{air} V_{disc}^2 A_p - \cancel{\rho_p V_{disc}} g$$

$$V_{disc} \sqrt{\frac{\rho_p V_{disc} g}{10.2 \rho_{air} A_p}} \Rightarrow \sqrt{\frac{\rho_p A_p h_{disc} g}{10.2 \rho_{air} A_p}} \quad \left[\frac{\rho_p V_{disc} g}{10.2 \rho_{air} A_p} \right]^{1/2}$$

$$V_{disc} = \sqrt{\frac{\rho_p h_{disc} g}{10.2 \rho_{air}}} = U_{T, disc (i)}$$

(ii) $F_{net} = m_{disc} a = \Sigma F = F_D - F_g$
 $= m_{disc} \frac{dV}{dt} = F_D - F_g$

$$m_{disc} = \rho_p V_{disc} \quad F_g = \rho_p V_{disc} g \quad F_D = C_{d, disc} \cdot \frac{1}{2} \rho_{air} V_{disc}^2 A_p$$

Assume @ low speeds, $Re = 1$ $C_{d, disc} = 13.6/Re \Rightarrow 13.6$

@ Terminal Velocity $\frac{dV}{dt} = 0 \Rightarrow$ Forces in Balance

$$\cancel{\rho_p V_{disc}} \cdot \frac{dV}{dt} = C_{d, disc} \cdot \frac{1}{2} \rho_{air} V_{disc}^2 A_p - \cancel{\rho_p V_{disc}} g$$

$$V_{disc} \sqrt{\frac{\rho_p V_{disc} g}{6.8 \rho_{air} A_p}} \Rightarrow \sqrt{\frac{\rho_p A_p h_{disc} g}{6.8 \rho_{air} A_p}} \quad \left[\frac{\rho_p V_{disc} g}{6.8 \rho_{air} A_p} \right]^{1/2}$$

$$V_{disc} = \sqrt{\frac{\rho_p h_{disc} g}{6.8 \rho_{air}}} = U_{T, disc (ii)}$$

Numerical

$$U_{T, Disc(i)} = \sqrt{\frac{F_p \text{ hoise } g}{10.2 \rho_{Air}}}$$

$$= \sqrt{\frac{(1000 \text{ kg/m}^3)(5.0 \times 10^{-6} \text{ m})(9.81 \text{ m/s}^2)}{10.2 (1 \text{ kg/m}^3)}}$$

$$= 0.069345 \text{ m/s} \approx 6.935 \times 10^{-2} \text{ m/s}$$

$$U_{T, Disc(ii)} = \sqrt{\frac{F_p \text{ hoise } g}{6.8 \rho_{Air}}}$$

$$= \sqrt{\frac{(1000 \text{ kg/m}^3)(5.0 \times 10^{-6} \text{ m})(9.81 \text{ m/s}^2)}{6.8 (1 \text{ kg/m}^3)}}$$

$$= 0.0849307 \text{ m/s} \approx 8.4931 \times 10^{-2} \text{ m/s}$$

Sphere

$$F_{Net} = M_{sphere} a = \Sigma F = F_D - F_g$$

$$= M_{Disc} \frac{dV}{dt} = F_D - F_g$$

$$M_{Disc} = \rho_p V_{sphere} \quad F_g = \rho_p V_{sphere} g \quad F_D = C_{d,sphere} \cdot \frac{1}{2} \rho_{Air} V_{sphere}^2 A_p$$

$$\Rightarrow \rho_p V_{sphere} \frac{dV}{dt} = C_{d,sphere} \cdot \frac{1}{2} \rho_{Air} V_{sphere}^2 A_p - \rho_p V_{sphere} g$$

$$V_{sphere} = \frac{\pi D^3}{6} \quad @ \text{ low speeds Assume } Re = 1 \Rightarrow C_{d,sphere} = 24.0$$

$$@ \text{ Terminal Velocity } \frac{dV}{dt} = 0$$

$$\Rightarrow 0 = 12 \rho_{Air} V_{sphere}^2 \left(\frac{\pi D^2}{4} \right) - \rho_p \frac{\pi D^3}{6}$$

$$V_{sphere} = \sqrt{\frac{\rho_p \frac{\pi D^3}{6} g}{12 \rho_{Air} \left(\frac{\pi D^2}{4} \right)}} = \sqrt{\frac{\rho_p D g}{18 \rho_{Air}}}$$

$$U_{T,sphere} = \sqrt{\frac{\rho_p D_{sphere} g}{18 \rho_{Air}}}$$

$$\text{For } U_{T,sphere} = U_{T,Disc(i)}$$

$$D_{sphere} = \frac{(U_{T,Disc(i)})^2 18 \rho_{Air}}{\rho_p g}$$

$$D_{sphere} = \frac{(0.069345 \text{ m})^2 18 (1 \text{ kg/m}^3)}{(1000 \text{ kg/m}^3)(9.81 \text{ m/s}^2)} = 0.0000088234 \text{ m}$$

$$\approx 8.82 \mu\text{m}$$

$$\text{For } U_{T,sphere} = U_{T,Disc(ii)}$$

$$D_{sphere} = \frac{(U_{T,Disc(ii)})^2 18 \rho_{Air}}{\rho_p g}$$

$$D_{sphere} = \frac{(0.0849307 \text{ m})^2 18 (1 \text{ kg/m}^3)}{(1000 \text{ kg/m}^3)(9.81 \text{ m/s}^2)} = 0.000013735 \text{ m}$$

$$\approx 13.24 \mu\text{m}$$